

CliniScan: AI-Based Chest X-Ray Abnormality Classification and Detection

1. Project Description

CliniScan is an AI-based medical imaging system designed to automatically analyse chest X-ray images and detect abnormalities using deep learning techniques. Chest X-rays are widely used in diagnosing lung diseases such as pneumonia, pulmonary fibrosis, and pleural effusion.

The goal of this project is to build an intelligent system capable of automatically identifying abnormalities in chest X-ray images. The system combines two deep learning approaches:

1. Multi-Label Classification – determines which diseases are present in the X-ray.
2. Object Detection – identifies the location of abnormalities using bounding boxes.

By combining classification and detection, the system provides both diagnostic prediction and spatial localization of disease regions.

2. Dataset Used

The dataset consists of 5000 chest X-ray images with corresponding annotations describing abnormalities present in each image.

Each annotation includes:

- image_id
- class_name
- class_id
- x_min, y_min
- x_max, y_max

These coordinates define bounding boxes that indicate the location of abnormalities within the X-ray image.

The dataset contains 15 pathology classes plus a background class for detection, giving a total of 16 classes.

3. Environment Setup

The system was implemented using Python and the PyTorch deep learning framework.

Libraries used:

- PyTorch (torch, torchvision) – deep learning models
- NumPy – numerical computations
- Pandas – dataset handling
- Matplotlib – visualization of training results

- Scikit-learn – evaluation metrics
- Pillow (PIL) – image loading
- tqdm – training progress visualization

The project was developed in Visual Studio Code and training was executed on CPU.

4. Data Exploration

Before training the model, the dataset was explored to verify structure and ensure data integrity.

Steps performed:

- Verified image files and annotation entries.
- Checked bounding box formats and values.
- Visualized sample images to ensure correct resolution.
- Removed invalid bounding boxes where width or height was too small.

These checks ensure reliable training data for the deep learning models.

5. Data Preprocessing

Several preprocessing steps were applied before training.

Image Loading:

Images are loaded using the Pillow library and converted to RGB format.

Normalization:

Pixel values are scaled to the range 0–1 to stabilize training.

Tensor Conversion:

Images are converted to PyTorch tensors with shape $C \times H \times W$.

MONOCHROME1 Correction:

Some X-ray images may have inverted brightness values. The system detects and corrects this automatically to ensure consistent medical representation.

6. Data Augmentation

Medical-safe augmentations were applied to improve model generalization.

Applied augmentations:

- Horizontal flipping
- Small rotations ($\pm 10^\circ$)
- Brightness adjustment
- Contrast adjustment
- Mild Gaussian noise
- Random scaling ($\pm 10\%$)

7. Proposed System

The CliniScan system consists of two major modules.

Multi-Label Classification Model

The classification module uses EfficientNet-B0 as the backbone network. EfficientNet is a convolutional neural network designed to achieve high accuracy while maintaining low computational cost.

In the implemented code, a pretrained EfficientNet-B0 model is loaded using transfer learning. The pretrained weights were originally trained on large image datasets, allowing the model to already understand basic visual patterns such as edges, textures, and shapes.

Most of the layers in the network are frozen, meaning their parameters are not updated during training. Only the final feature block and the classifier layer are trained. This allows the model to adapt specifically to chest X-ray images while retaining the general image features learned from pretraining.

The final classifier layer is replaced with a custom fully connected layer that outputs predictions for multiple disease classes. Since a single chest X-ray can contain more than one abnormality, the model performs multi-label classification, where each disease is predicted independently.

The model identifies abnormalities by extracting hierarchical visual features from the X-ray image and mapping them to disease probabilities.

EfficientNet-B0 was chosen because it provides a good balance between accuracy, efficiency, and computational cost, making it suitable for medical image classification tasks.

Abnormality Detection Model

For locating abnormalities within the X-ray images, the system uses Faster R-CNN with a ResNet-50 and Feature Pyramid Network.

Unlike classification models that only determine whether a disease exists, object detection models also identify where the abnormality is located in the image.

The Faster R-CNN architecture works in two main stages:

1. Feature Extraction
The ResNet-50 backbone extracts deep feature maps from the chest X-ray image.
2. Region Proposal and Detection
A Region Proposal Network scans the feature maps and suggests possible regions that may contain abnormalities. These regions are then classified into disease categories and refined to produce accurate bounding boxes.

The Feature Pyramid Network (FPN) helps detect abnormalities at multiple scales, allowing the model to identify both small nodules and larger lung opacities.

Faster R-CNN was chosen because it provides high detection accuracy and precise localization, which is critical for medical imaging applications where identifying the exact position of abnormalities is important.